DP Biology – B3.2 (HL, Plant Systems)

Root pressure and phloem

- Which adaptations help animals and plants move fluids around?
- How does transport in animals compare with transport in plants?

HL statements B3.2.17–18. Textbook pp. 321–325.



Image: part of a sunflower stem, stained section. Jon Houseman, CC BY-SA 4.0.

Objectives for each class

Class	Statements	By the end of the class, you can...
1	B3.2.17–18	Explain how root pressure is generated, and how phloem is adapted for translocation

Root pressure

B3.2.17



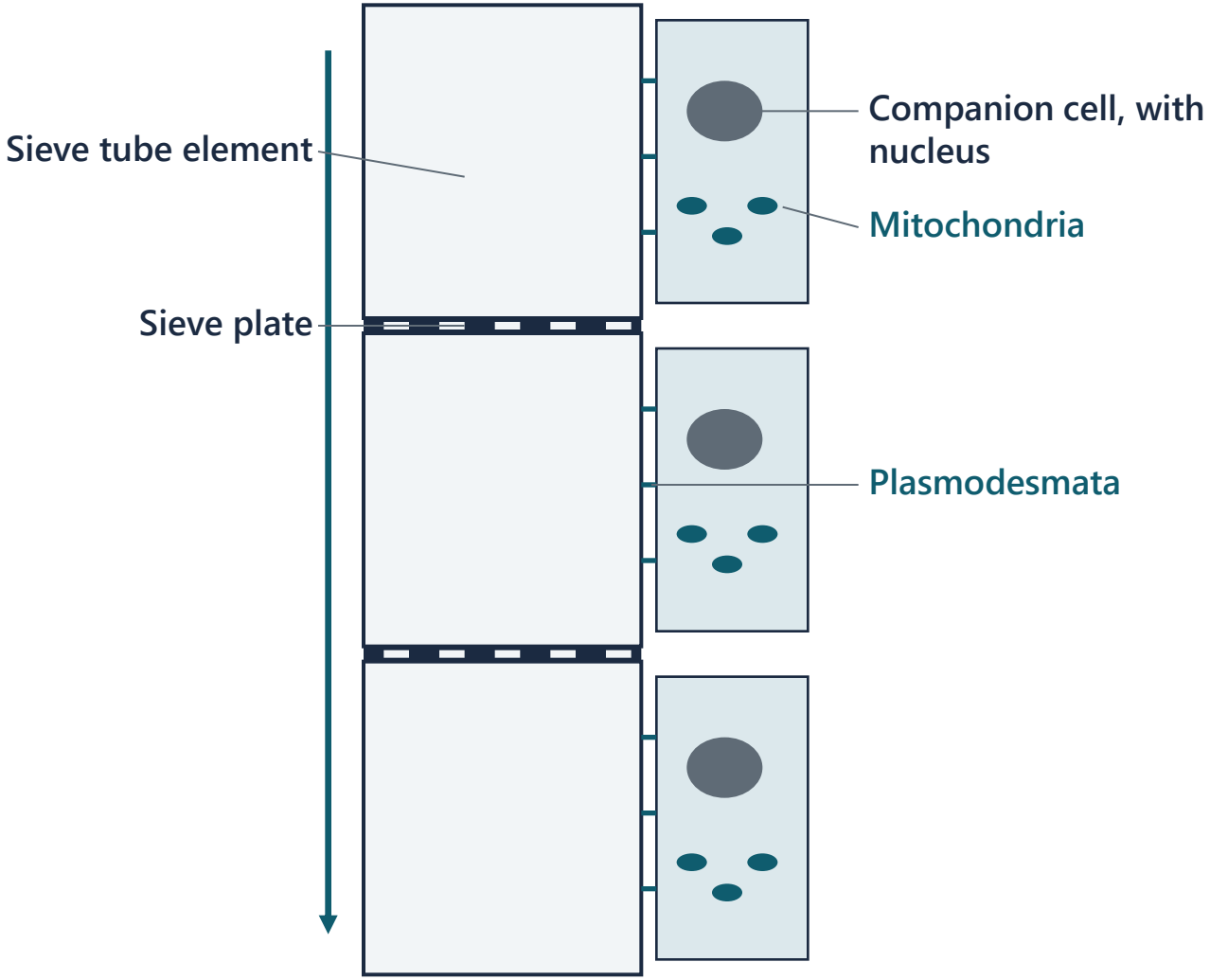
Guttation: drops pushed out of a leaf by root pressure. Public domain.

- Usually water is **pulled** up by transpiration (tension)
- When transpiration is low, roots can **push** sap up instead
- This happens in very humid air, at night, or in spring before deciduous leaves open
- It can refill xylem vessels that have filled with air

Root cells pump mineral ions into the xylem (active transport, ATP) → Xylem sap becomes more concentrated → Water follows by osmosis → Positive pressure pushes sap up the xylem

Sieve tubes and companion cells

B3.2.18



Simplified longitudinal section. The arrow shows sap flow, which can go either way.

- **Translocation**: moving sap with sugars and amino acids from **sources** to **sinks**
- Source: where sugars are made or released, such as mature leaves
- Sink: where they are used or stored, such as roots, fruits and growing buds
- Sap flows under pressure in living sieve tubes

Adaptations for translocation

B3.2.18

Feature	How it helps
Sieve plates with large pores	Sap flows from one element to the next
Little cytoplasm, few organelles	Less resistance to the flow of sap
No nucleus in sieve tube elements	More space for flow; the companion cell keeps the element alive
Many mitochondria in companion cells	ATP for active loading of sucrose at sources
Plasmodesmata between the two cells	Sucrose and other molecules pass into the sieve tube
Thick, rigid cell walls	Withstand the high pressure of the sap

Key idea

Loading at sources raises pressure; unloading at sinks lowers it, so sap flows from source to sink.

Check that you can explain

1. How active transport of mineral ions in roots generates root pressure
2. When root pressure matters more than the pull from transpiration
3. How sieve tube elements and companion cells are adapted
4. Why sap flows from sources to sinks

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Outline how active transport in root cells leads to root pressure.

2. State two adaptations of sieve tube elements for the flow of sap.

3. Explain why sap flows from a source to a sink.

Image credits

Helianthus stem.jpg – Jon Houseman – CC BY-SA 4.0

Guttation ne.jpg – NoahElhardt assumed (ba – Public domain

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